

further increased by injecting multiple signals of slightly differing wavelengths. This is called wavelength-division multiplexing (WDM) and is an important technique to increase the system bandwidth," explains Taws.

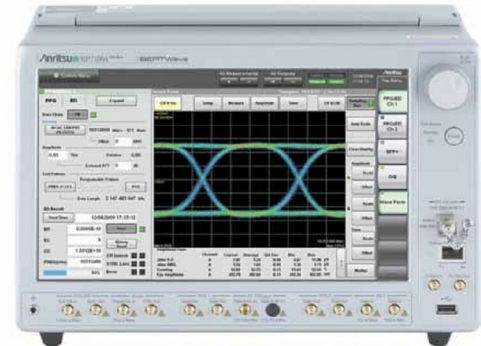
Fibre-optic networking is more than a decade old and existed along with synchronous optical networking (SONET) and synchronous digital hierarchy (SDH), which today goes up to over 10 GB.

In one of the interviews, Kumar N. Sivarajan, co-founder and chief technology officer, Tejas Networks, states that the recent development is dense wavelength-division multiplexing (DWDM), which includes the use of multiple frequencies or wavelengths for different signals from the infrared light portion of the optical spectrum. Today, optical networking has become synonymous with DWDM.

Optical testing challenges

Optical communications is a fairly stable technology with established processes and best practices. Therefore the initial ramp-up needed for manpower to operate this technology is also well-defined and easily available.

However, optical communication faces other types of issues. Madhukar Tripathi, manager-telecom segment, Anritsu, shares, "The major disadvantage is that it is very expensive to construct the fibre-optics for optical networks and to join the fibre-optical



Anritsu's BERTWave bit-error rate and eye-pattern analyser

cables as compared to the copper cables."

Mombasawala says, "The laying of optical cables, securing rights-of-way, frequent maintenance due to dig-outs and the necessity of the backup routes for network resiliency are some of the challenges faced in optical networks."

"The testing of optical networks during installation and commissioning includes link verification, link characterisation, system tests like bit error rate (BER) measurements, alarms and transmission power measurements. This type of testing is very repetitive and needs standard tools like optical-time domain reflectometer (OTDR), BER meter and power meter, while R&D tests demand sound technical knowledge and skills to test the systems," he adds.

Expressing his views on the challenges faced in India, Taws says, "One of the main challenges we have in India is ensuring that the optical connections are free of dust and dirt during installation and maintenance. Contamination is the main reason for trouble in optical networks. A single particle mated into the core of a fibre can cause significant back reflection, insertion loss and equipment damage. Visual inspection is the only way to

determine whether fibre connectors are truly clean before mating them."

Key measurements in optical communication

In order to qualify an optical fibre or an optical fibre system for proper transmission, several key measurements are performed. "Critical measurements such as OTDR, chromatic dispersion (CD), polarisation-mode dispersion (PMD), loss test set and optical return loss can be made quickly and accurately to maximise the network performance and accelerate deployment of new services," shares Tripathi. Many basic and advanced systems are available to perform these tests.

In newer systems designed for high-speed communications, the common test areas include coherent detection, which involves measurement of various link impairments such as CD, PMD loss and polarisation-dependent loss (PDL). Even the effect of non-linear link impairments can be qualified by using a technique based on error vector magnitude.

Mombasawala explains, "Optical modulation analysis includes tests like optical I-Q diagram, I-Q eye diagram, constellation diagram, EVM, phase-

Basic tests in optical communications

1. Inspection and cleaning
2. End-to-end optical link loss
3. Rate of attenuation per unit length
4. Attenuation contribution to splices, connectors and couplers (events)
5. Length of the fibre or distance to an event
6. Linearity of fibre loss per unit length (attenuation discontinuities)
7. Reflectance or optical return loss
8. Chromatic dispersion
9. Polarisation-mode dispersion
10. Attenuation profile